NATURAL LANGUAGE PROCESSING

Peer Review Analysis in Academia

Comparative Analysis Using ChatGPT and Google Gemini

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1. Introduction

Academic peer review is a process in which experts evaluate a research paper before it is published. Reviewers identify the strengths and weaknesses of the paper and suggest improvements to increase its quality.

Natural Language Processing (NLP) can be used to analyse peer reviews automatically. It can identify the overall tone, positive comments, negative comments, specific criticisms, and suggested improvements. It can also help distinguish constructive criticism from personal or disrespectful language.

In this assignment, two AI tools, ChatGPT and Google Gemini, are used to analyse the same academic peer review. Their outputs are compared to understand how effectively they identify the tone and content of the review.

1.1 Objectives

- To analyse the overall tone of an academic peer review.
- To identify positive comments and negative comments.
- To extract specific criticisms and suggested improvements.
- To distinguish constructive criticism from personal or disrespectful language.
- To compare the outputs of ChatGPT and Google Gemini.
- To identify limitations and discuss the adaptation of peer-review analysis for Hindi and Hinglish.

2. Input Data

The following academic peer review was used as input for both tools.

Sample Academic Peer Review:

The paper addresses an important problem, and the proposed method is interesting. However, the dataset is too small, the baseline comparisons are weak, and the methodology is not

explained clearly. The authors should include more experiments and improve the discussion of limitations. The paper has potential, but substantial revisions are required before it can be considered for publication.

The same text was submitted to ChatGPT and Google Gemini to ensure a fair comparison.

3. Tool 1: ChatGPT

Tool name: ChatGPT

Website: https://chatgpt.com/

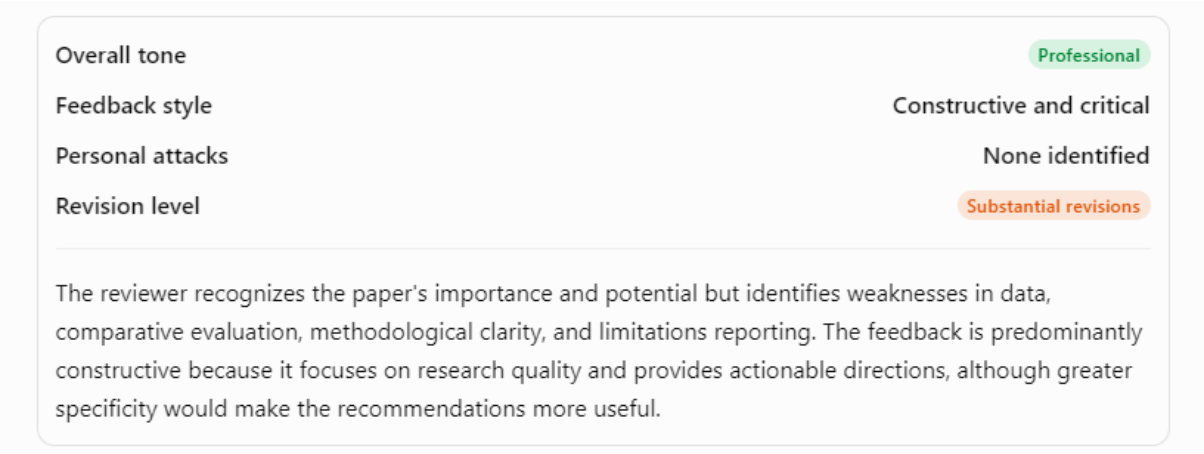
Purpose: To analyse the tone, positive comments, negative comments, specific criticisms, and suggested improvements in the academic peer review.

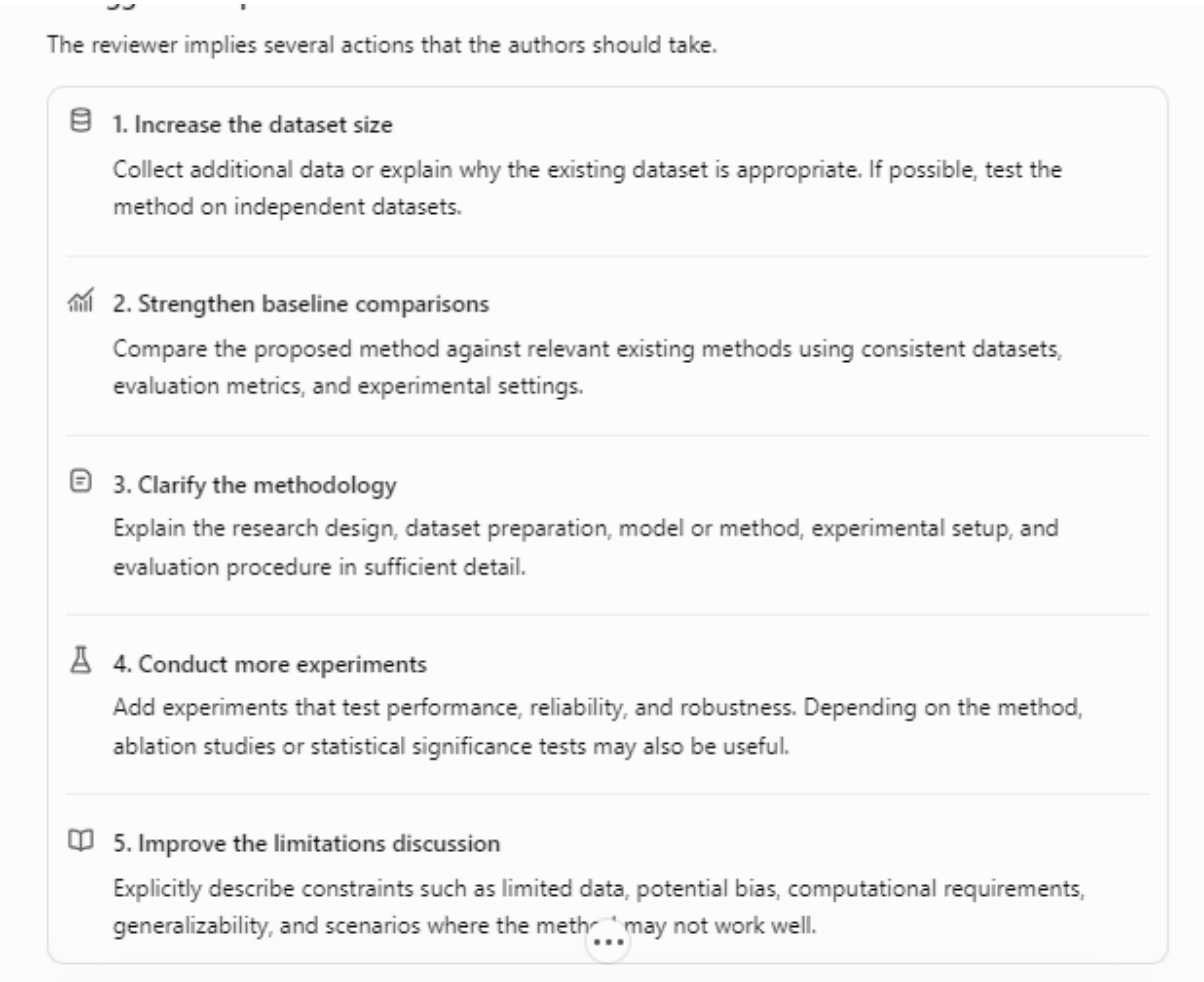
3.1 Input Prompt

The following prompt was used:

“Analyse this academic peer review. Identify its overall tone, positive comments, negative comments, specific criticisms, and suggested improvements. Distinguish constructive criticism from personal or disrespectful language.”

3.2 Output Analysis





A. Overall Tone

The overall tone is professional, constructive, and moderately critical.

The reviewer recognises the importance of the research problem and finds the proposed method interesting. However, the reviewer identifies weaknesses that must be addressed before publication.

B. Positive Comments

- The paper addresses an important research problem.
- The proposed method is interesting.
- The paper has potential for publication after substantial revisions.

C. Negative Comments

- The dataset is too small.
- The baseline comparisons are weak.
- The methodology is not explained clearly.
- More experiments are required.
- The limitations discussion needs improvement.

D. Suggested Improvements

- Increase the dataset size or justify why the current dataset is sufficient.
- Compare the proposed method with stronger baseline approaches.
- Explain the methodology more clearly.
- Conduct additional experiments.
- Discuss the limitations of the research in greater detail.

E. Constructive Criticism vs. Disrespect

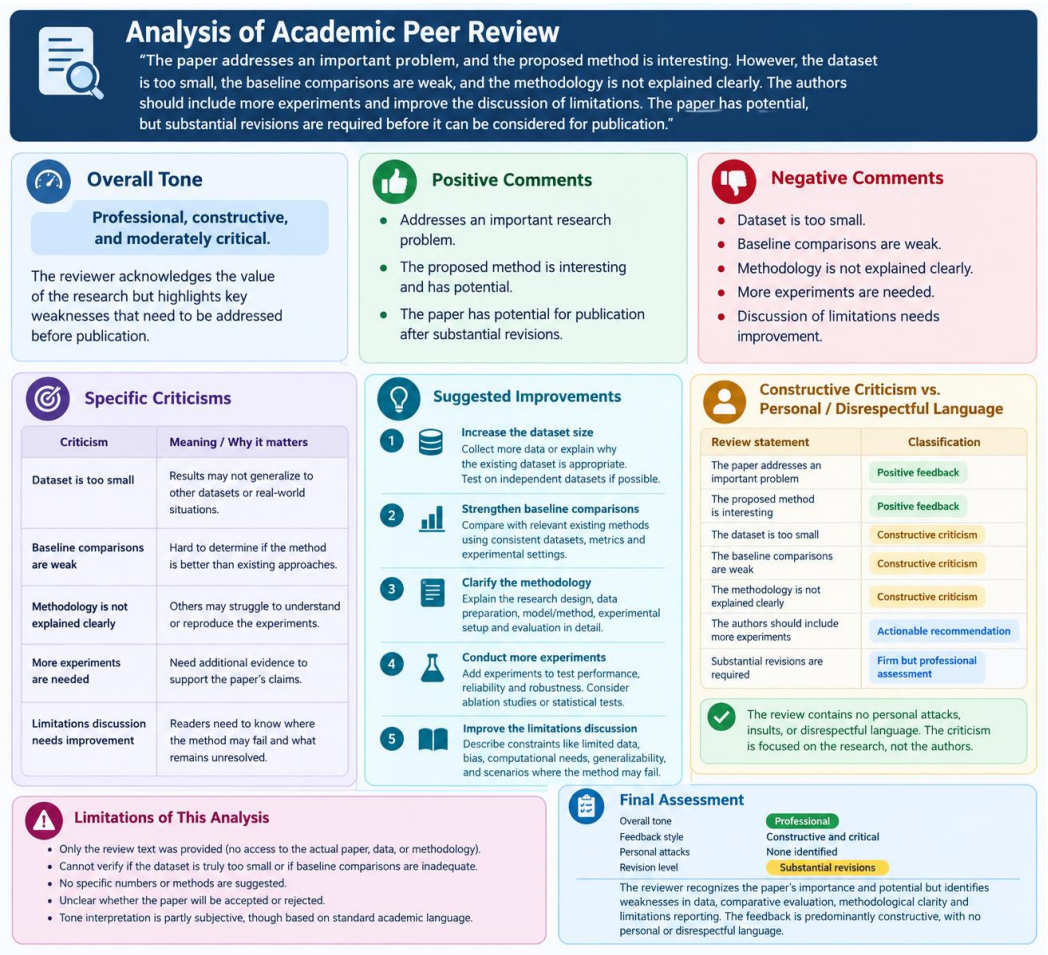
ChatGPT identifies the criticism as research-focused and constructive. The reviewer discusses the dataset, methodology, and experimental evaluation rather than attacking the authors personally.

No explicit insults, mocking, or personal attacks are identified.

F. Limitations

The analysis is based only on the review text. ChatGPT cannot independently verify whether the dataset is actually too small or the baseline comparisons are inadequate without examining the original paper.

Screenshot 1:



4. Tool 2: Google Gemini

Tool name: Google Gemini

Website: https://gemini.google.com/

Purpose: To analyse the same academic peer review and identify its tone, strengths, weaknesses, and recommendations.

4.1 Input Prompt

The same prompt and academic peer-review text were used in Google Gemini.

“Analyse this academic peer review. Identify its overall tone, positive comments, negative comments, specific criticisms, and suggested improvements. Distinguish constructive criticism from personal or disrespectful language.”

4.2 Output Analysis

A. Overall Tone

The overall tone is professional, constructive, and moderately critical.

The reviewer acknowledges that the paper addresses an important problem and that the proposed method is interesting. However, several weaknesses need to be addressed before the paper can be considered for publication.

B. Positive Comments

- The research problem is important and relevant.
- The proposed method is interesting.
- The paper has potential after substantial revisions.

C. Negative Comments

- The dataset is insufficiently large.
- The baseline comparisons are weak.
- The methodology lacks clarity.
- Additional experiments are needed.
- The discussion of limitations requires improvement.

D. Suggested Improvements

- Increase or justify the dataset size.
- Strengthen comparisons with existing approaches.
- Explain the methodology in greater detail.
- Conduct more experiments to support the research claims.

- Discuss the limitations and boundaries of the proposed method.

E. Constructive Criticism vs. Disrespect

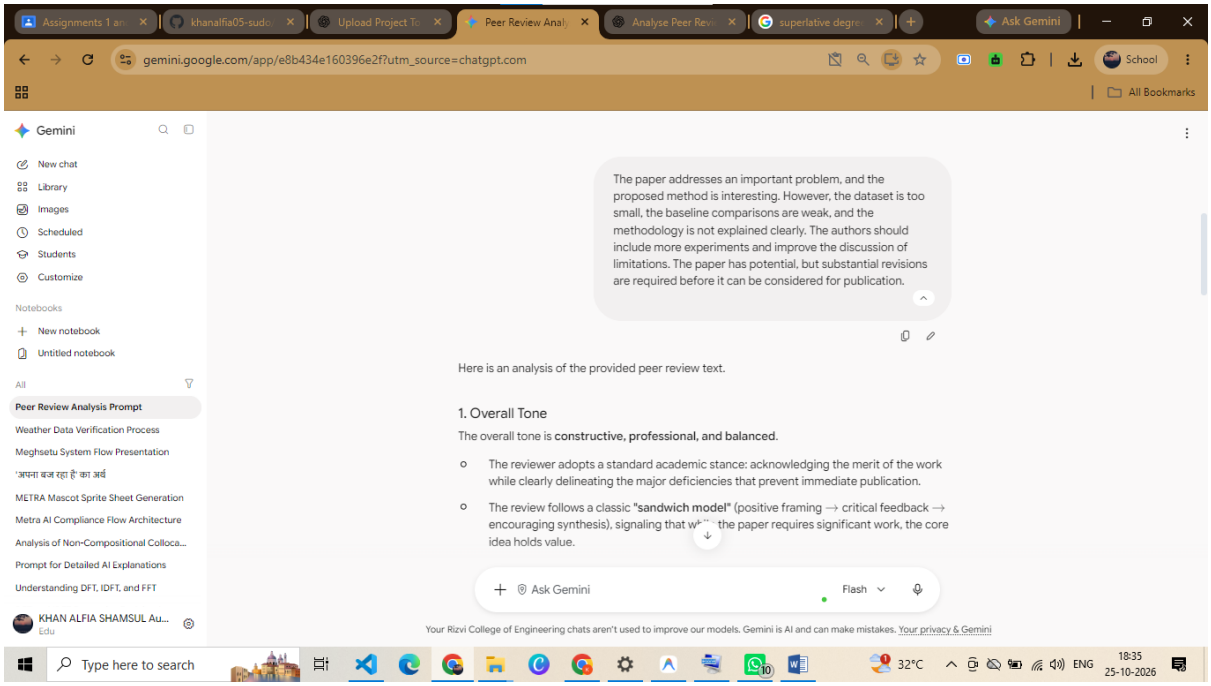
Gemini identifies the comments as constructive because they focus on the quality of the research. The reviewer provides reasons why improvements are needed without using explicit personal attacks or insulting language.

The statement that substantial revisions are required is firm but professional.

F. Limitations

Gemini cannot verify the technical accuracy of the reviewer's comments without access to the original research paper, experimental results, and methodology.

Screenshot 2:



5. Comparative Analysis of ChatGPT and Google Gemini

Both tools were tested using the same input text and a similar prompt. Their responses were compared according to several evaluation criteria.

Evaluation criterion	ChatGPT	Google Gemini
Overall tone	Professional, constructive, moderately critical	Professional, constructive, moderately critical

Positive comments	Identified the important problem, interesting method, and publication potential	Identified the same positive points
Negative comments	Identified the small dataset, weak baselines, and unclear methodology	Identified the same major criticisms
Suggested improvements	Recommended more data, stronger comparisons, clearer methodology, and additional experiments	Provided similar recommendations
Personal attacks	None identified	None identified
Limitations	Explained the lack of context and inability to verify technical claims	Identified similar limitations
Output structure	Detailed, with headings and explanations	Detailed, with headings and explanations

5.1 Findings

Both tools produced similar interpretations of the academic peer review. They identified the same major strengths and weaknesses and classified the tone as professional and constructive.

Both tools also recognised that the reviewer was criticising the research rather than the authors personally.

The main difference may lie in the wording, organisation, and level of explanation. Based on the outputs tested, neither tool demonstrated a clear advantage in identifying the review's main points.

6. Error Analysis Matrix

An error analysis matrix helps identify possible mistakes, missed information, and limitations in the tools' outputs.

Error or evaluation type	ChatGPT	Google Gemini
Incorrect tone classification	No obvious error	No obvious error
Missed positive comments	None apparent	None apparent
Missed major criticisms	None apparent	None apparent
Criticism incorrectly classified as personal abuse	No	No
Inferences beyond the review text	Some inferred implications, such as possible generalisation problems	Similar inferred implications
Formatting issues in the pasted output	Tables appeared broken	Tables appeared broken
Indian cultural misunderstanding	Not tested	Not tested

6.1 Interpretation of the Error Matrix

Both tools performed similarly on the standard-English review. Neither output showed an obvious mistake in identifying the overall tone or the major criticisms.

However, both tools inferred possible consequences, such as poor generalisability, from the statement that the dataset was too small. These are reasonable interpretations, but they cannot be confirmed without examining the original paper.

The cultural and regional evaluation remains incomplete because the input did not contain Hindi, Hinglish, regional expressions, or culturally specific language.

The results should therefore be treated as a qualitative comparison rather than a formal measurement of accuracy.

7. Constructive Criticism vs. Personal or Disrespectful Language

Constructive criticism identifies weaknesses in the work and suggests improvements. Disrespectful language attacks the authors personally instead of evaluating their research.

Review statement	Classification	Explanation
“The paper addresses an important problem.”	Positive feedback	Recognises the relevance of the research.
“The proposed method is interesting.”	Positive feedback	Acknowledges the potential value of the approach.
“The dataset is too small.”	Constructive criticism	Identifies a possible weakness in the data.
“The baseline comparisons are weak.”	Constructive criticism	Points out a concern with the evaluation.
“The methodology is not explained clearly.”	Constructive criticism	Identifies a clarity and reproducibility issue.
“The authors should include more experiments.”	Actionable recommendation	Suggests how the paper could be strengthened.
“Substantial revisions are required.”	Firm but professional assessment	Communicates that major improvements are necessary.

Finding: The supplied review contains no explicit personal attacks, insults, or mocking. The criticism is directed at the paper's methodology, data, and experimental evaluation.

8. Limitations of the Evaluation

The experiment has several limitations:

1. Limited input: Only one short academic peer review was tested.
2. Lack of context: The original research paper was not provided, so the technical criticisms could not be independently verified.

3. Qualitative comparison: The tools were compared by reading their outputs. No formal accuracy, precision, recall, or F1 score was calculated.
4. Language limitation: The test used standard academic English only.
5. No cultural evaluation yet: Hindi, Hinglish, and regional-language examples were not tested.
6. General-purpose tools: ChatGPT and Gemini are general-purpose AI assistants, not dedicated peer-review management systems.

These limitations mean that the results should not be generalised to every academic review or language.

9. Adaptation for Hindi and Hinglish

An NLP system designed for Indian academic peer reviews should be capable of understanding Hindi and Hinglish, including formal and informal expressions.

Hindi may be written in Devanagari script or Roman script. Hinglish combines Hindi and English words in the same sentence. These variations can make tone and sentiment analysis more challenging.

9.1 Suggested Adaptation Steps

1. Collect a suitable dataset: Create a dataset of academic review comments in Hindi, Hinglish, and English, with appropriate permissions.
2. Annotate the data: Label examples as positive feedback, constructive criticism, recommendations, disrespectful language, or neutral comments.
3. Build a bilingual glossary: Include academic terms such as dataset, baseline, methodology, experiment, bias, and limitation, along with their Hindi equivalents and common mixed-language forms.
4. Normalise text: Handle spelling variations, repeated characters, punctuation, and common Romanized Hindi forms while preserving meaningful tone.
5. Detect code-switching: Identify sentences containing a mixture of Hindi and English.
6. Use multilingual models: Evaluate language models that support Hindi and English, then compare their results with human annotations.
7. Conduct human evaluation: Ask Hindi-speaking reviewers to check ambiguous comments, sarcasm, indirect criticism, and potentially disrespectful expressions.
8. Test different language varieties: Evaluate Devanagari Hindi, Romanized Hindi, Hinglish, and regional variations separately.

9.2 Linguistic Resources Required

- A curated Hindi and Hinglish academic review dataset.
- A Hindi-English academic terminology glossary.
- Hindi text normalisation and tokenisation resources.
- Annotated examples of praise, criticism, recommendations, sarcasm, and personal attacks.
- Human evaluators familiar with Hindi, Hinglish, and academic writing conventions.

9.3 Sociolinguistic Challenges

Challenge	Explanation	Possible solution
Code-switching	Hindi and English may appear in the same sentence.	Support mixed-language text analysis.
Script variation	Hindi may be written in Devanagari or Roman script.	Train and test the system on both scripts.
Regional dialects	Vocabulary and expressions vary across regions.	Include diverse examples and regional reviewers.
Indirect criticism	A reviewer may soften negative feedback to sound polite.	Analyse context rather than relying only on negative keywords.
Academic terminology	English research terms may remain untranslated in Hindi sentences.	Create a domain-specific bilingual glossary.
Sarcasm and ambiguity	A sentence may sound positive but imply criticism.	Use contextual analysis and human review for uncertain cases.

9.4 Example of Hinglish Peer Review

Input:

“Paper ka idea achha hai, lekin dataset ka size kaafi small hai aur methodology clearly explain nahi ki gayi hai. Authors ko aur experiments karne chahiye.”

Expected analysis:

- Overall tone: Constructive but critical.
- Positive comment: The research idea is good.
- Negative comments: The dataset is small, and the methodology is unclear.
- Suggested improvement: Conduct more experiments.
- Personal attacks: None apparent.

This is an illustrative expected analysis, not an observed result from ChatGPT or Gemini. To evaluate cultural and regional performance, enter this example into both tools and record their actual outputs.

10. Ethical Considerations

Academic peer reviews may contain confidential information about unpublished research. Therefore, several ethical considerations are important:

- Confidential review text should not be uploaded to external AI tools unless permitted by the relevant institution or publisher.
- Personal information and confidential manuscript details should be protected.
- AI-generated results should support human judgement rather than replace expert reviewers or editors.
- Direct criticism should not automatically be classified as disrespectful.

- Systems should be evaluated for bias across languages, dialects, and writing styles.
- Uncertain classifications should be flagged for human review rather than treated as definite conclusions.

11. Conclusion

This assignment analysed an academic peer review using ChatGPT and Google Gemini. Both tools identified the review as professional, constructive, and moderately critical.

They recognised the positive aspects of the research, including the importance of the problem and the potential of the proposed method. They also identified the main weaknesses: a small dataset, weak baseline comparisons, unclear methodology, insufficient experiments, and an inadequate discussion of limitations.

Both tools suggested similar improvements and found no explicit personal attacks or disrespectful language.

However, the evaluation was limited to one English-language review, and the original paper was not available for independent verification. Therefore, the results demonstrate similarity between the two outputs rather than proving that either tool is more accurate.

A Hindi and Hinglish adaptation would require suitable linguistic resources, multilingual models, support for script variation and code-switching, and human evaluation. Such adaptation could make peer-review analysis more useful for researchers working in India's multilingual academic environment.

12. References

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